Computer Vision

Blood Cell Object Detection: Fine-Tuning Yolo Variants

In this experiment, I trained and evaluated two YOLOv8 variants: YOLOv8-Nano and YOLOv8-Medium using a Tesla T4 GPU.

Variant 1: YOLOv8 Nano

Performance metrics:

Mean Average Precision (mAP) @ 0.50 IoU threshold	0.9270
mAP @ 0.50-0.95 IoU threshold	0.6412
Recall	0.9184
Precision	0.8555
Inference Speed	5.3 ms
Model Size	~6.2 MB

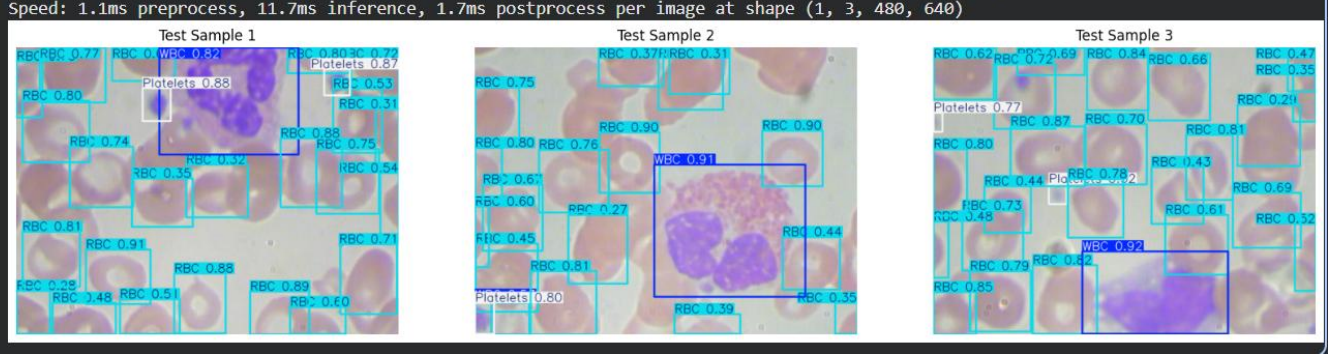
Class-by-Class Analysis:

Class	Precision	Recall	mAP50	Interpretation
WBC	0.973	1.00	0.995	Perfect. White Blood Cells are huge and distinct. The model practically never misses them (Recall=1)
RBC	0.787	0.815	0.865	The Hardest Class. Red Blood Cells overlap heavily. The lower recall, some RBCs are missed (crowd).
Platelets	0.807	0.940	0.922	High Detection despite them being small. However, the lower precision (0.80) means it sometimes thinks a speck of dust is a platelet.

```
...
image 1/1 /content/2022170413/yolo_dataset/images/test/BloodImage_00223.jpg: 480x640 1 WBC, 23 RBCs, 2 Plateletss, 65.9ms
Speed: 1.6ms preprocess, 65.9ms inference, 1.4ms postprocess per image at shape (1, 3, 480, 640)

image 1/1 /content/2022170413/yolo_dataset/images/test/BloodImage_00251.jpg: 480x640 2 WBCs, 17 RBCs, 1 Platelets, 7.5ms
Speed: 2.3ms preprocess, 7.5ms inference, 1.6ms postprocess per image at shape (1, 3, 480, 640)

image 1/1 /content/2022170413/yolo_dataset/images/test/BloodImage_00335.jpg: 480x640 1 WBC, 23 RBCs, 2 Plateletss, 11.7ms
Speed: 1.1ms preprocess, 11.7ms inference, 1.7ms postprocess per image at shape (1, 3, 480, 640)
```



Variant 2: YOLOv8 Medium

Performance metrics:

Mean Average Precision (mAP) @ 0.50 IoU threshold	0.9228
mAP @ 0.50-0.95 IoU threshold	0.6394
Recall	0.8916
Precision	0.8501
Inference Speed	15.7 ms
Model Size	~52.0 MB

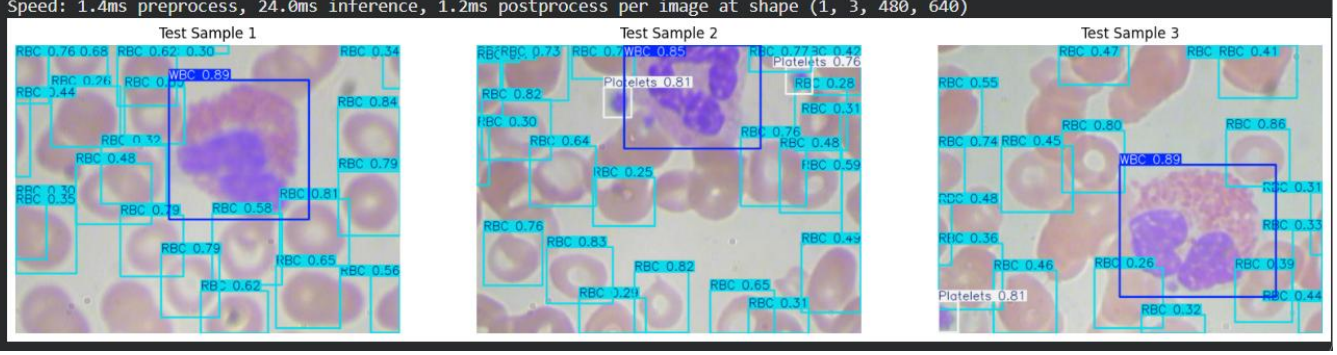
Class-by-Class Analysis:

Class	Precision	Recall	mAP50	Interpretation
WBC	0.956	1.00	0.994	Excellent. (Recall=1), though slightly less precise than the Nano model (more false positives)
RBC	0.765	0.803	0.862	Performance dropped. The larger model struggled more with the dense overlaps, missing nearly 20% of RBCs
Platelets	0.829	0.871	0.912	Good precision, lower recall. It was more careful than the Nano model (higher precision) but missed significantly more platelets

```
image 1/1 /content/2022170413/yolo_dataset/images/test/BloodImage_00289.jpg: 480x640 1 WBC, 21 RBCs, 41.0ms
Speed: 1.5ms preprocess, 41.0ms inference, 1.3ms postprocess per image at shape (1, 3, 480, 640)

image 1/1 /content/2022170413/yolo_dataset/images/test/BloodImage_00223.jpg: 480x640 1 WBC, 22 RBCs, 2 Plateletss, 27.6ms
Speed: 1.4ms preprocess, 27.6ms inference, 1.2ms postprocess per image at shape (1, 3, 480, 640)

image 1/1 /content/2022170413/yolo_dataset/images/test/BloodImage_00251.jpg: 480x640 1 WBC, 17 RBCs, 1 Platelets, 24.0ms
Speed: 1.4ms preprocess, 24.0ms inference, 1.2ms postprocess per image at shape (1, 3, 480, 640)
```



And the following explain how the models performed during training to get such results. The training process was monitored over 50 epochs. The logs reveal several key behavioral patterns:

1.For YOLOv8-Nano:

- Slower Initial Learning: due to its few parameters (3.2M vs 25.9M), it took slightly longer to grasp the initial features. It needed about 10 epochs to reach the high-performance zone.
- Consistent Improvement (No Overfitting): steadiness indicates that it was genuinely learning generalizable features rather than just memorizing the training set.
- Training Time: 0.074 hours (~4.4 minutes)
- Consumed only ~3.0 GB of VRAM
- Higher final mAP (0.927)

2.For YOLOv8-Medium:

- Rapid Convergence: The model achieved 97% of its final accuracy within the first 20% of the training time, allowing it to memorize features almost instantly.
- Loss Minimization The internal error metrics showed steady improvement without signs of divergence (instability):
  - Box Loss (Localization Error): Decreased from 1.471 (Epoch 1) to 0.741 (Epoch 50). This means the model got progressively better at drawing tight boxes around the cells.
  - Classification Loss (Identification Error): Decreased drastically from 2.209 (Epoch 1) to 0.392 (Epoch 50). The model quickly learned to distinguish purple WBCs from red RBCs.
- The model stopped gaining significant accuracy halfway through the training. Continuing to train beyond 30-40 epochs yielded diminishing returns, confirming that the model had fully saturated the information available in the training set.
- Training Time: 0.171 hours (~10.2 minutes)
- GPU Memory Usage: Peaked at 7.4 GB.

Comparison: While 10 minutes is fast, the Nano model (Model A) likely completed this same task in 3-4 minutes with significantly less VRAM usage.